

Relax Inc Take-home Challenge

By Brad Wicklund

The goal of this project was to define “adopted users” as those who logged in on at least three distinct days within a seven-day window, and to identify which factors best predict adoption. I combined the `takehome_users` and `takehome_user_engagement` datasets, converted key datetime fields, and engineered features such as account age, days since last session, and whether a user was invited by another. I then created an adoption flag by scanning each user’s login history and merged this target variable back into the user table for analysis.

Exploratory analysis showed that the median user logged in only once, with the 90th percentile reaching 53 active days. Overall adoption was low, at around 13–15 percent of users. Adoption varied meaningfully by signup source: users signing up through Google authentication and those invited by others were more likely to adopt, while organizational invites and personal project accounts were much less likely to do so. Marketing features such as mailing list opt-in and drip campaign enrollment showed little predictive power, suggesting that communication alone does not drive adoption.

To quantify these effects, I fit a logistic regression model. An initial model that included a “has session” flag achieved near-perfect performance, but this was deemed a leakage variable. After removing it, the final model achieved a ROC-AUC of 0.87 and accuracy of 0.86, with the strongest predictors being signup source and invitation status. Account age and session recency had only marginal effects, and marketing flags remained weak.

In summary, adoption is best explained by how a user joins the product, not by subsequent marketing outreach. Future work could investigate organizational size, email domain effects, or cohort trends, and test tree-based models to capture more complex patterns.

References

[https://github.com/bradwicklund/Springboard/blob/main/Take Home Challenges/wicklund_relax_challenge/1.0-bjw-relax-inc.ipynb](https://github.com/bradwicklund/Springboard/blob/main/Take%20Home%20Challenges/wicklund_relax_challenge/1.0-bjw-relax-inc.ipynb)